

Bareksa: Helping Indonesians grow their wealth through investments with Google Kubernetes Engine

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ABOUT BAREKSA



About Bareksa

Bareksa grew as a management by slashed database management time freed up the DevOps to pursue fintech and forged an ML

Bareksa is an Indonesian fintech on a mission to spread the power of investment to millions of Indonesians who are joining the middle classes. The startup's digital platform offers an online mutual fund marketplace and robo advisory service that opens access to a wealth of investment opportunities so Indonesians can grow wealth and plan for a secure future. In 2021 Bareksa won Series C investment from ride-hailing giant and super-app Grab.

for curated service Google Cloud.

Industries: Financial Services & Insurance

Location: Indonesia

Google Cloud results

- Liberates development team to pursue creative tasks by slashing database management tasks by 60%
- Spurs major platform growth by improving customer satisfaction rating from 3.8 to 4.7
- Boosts productivity by eliminating lost files due to system outages

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asset
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(<https://workspace.google.com/>), Docs

Google Cloud (<https://cloud.google.com/>) /docs/

BigQuery (<https://cloud.google.com/bigquery/>)

(<https://workspace.google.com/products/drive/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Sheets

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Google Workspace

(<https://workspace.google.com/>)

Docs

(<https://workspace.google.com/products/docs/>)

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Google Drive

(<https://workspace.google.com/products/drive/>)

Sheets

(<https://workspace.google.com/products/sheets/>)

through
Google
Workspace
migration

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—*Trio Purnomo, CTO,*
Bareksa

Indonesia is one of Asia's most promising economies, with a huge, relatively young population of 270 million rapidly embracing digital technologies and building their wealth. According to the World Bank, the number of middle class (<https://www.worldbank.org/en/country/indonesia/publication/aspiring-indonesia-expanding-the-middle-class>) Indonesians has grown from 2002 at a rate of roughly 10% each year to reach 52 million in 2020.

Amid the impressive rate of wealth generation in the country, millions of newly minted middle class Indonesians still lack the tools and knowledge to plan for a secure future through investment. Fintech startup Bareksa (<https://www.bareksa.com/>) was founded in 2013 on a mission to fill this gap. It is an online mutual fund marketplace and financial robo advisory that lets anybody with a mobile device access the opportunity to invest.

Bareksa started life in a monolithic on-premise environment, with physical servers and a self-managed database. As the popularity of the platform grew, it found itself struggling to scale services when hit with a spike in customer demand, a big problem for a startup that relied on speed and 100% reliability to succeed. Meanwhile, its engineering team was spending so much time on DevOps tasks, such as provisioning servers and managing databases, that it had little time for the key mission of creating innovative fintech solutions to grow the business.

The startup decided to embark upon a comprehensive migration to Google Cloud

(<https://cloud.google.com/>), adopting Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) (GKE) as its scalable infrastructure, and Cloud SQL

(<https://cloud.google.com/sql>) as its relational database. With this move, Bareksa was able to grow its assets under management (AUM) by five times, in just one year.

"Our on-prem architecture held us back from growing at the pace of our ambitions," says Trio Purnomo, CTO, Bareksa. "By switching to Google Kubernetes Engine, we gained an unlimited ability to scale our solution and unleash new innovation."

Powering fintech innovation to boost customer satisfaction with Google Kubernetes Engine

Bareksa operates with a lean team of only three DevOps engineers tasked with keeping the platform running 24/7, no matter how wild the market swings or fluctuations in customer demand. On an on-premise architecture, the Bareksa team was constantly scrambling to spin up and down virtual machines (VMs) for its growing number of financial services. They often needed to scale at least 20 service VMs just to keep pace with orders, a highly laborious, manual process.

By moving to GKE, Bareksa is now able to seamlessly autoscale compute power for its hundreds of services with just six nodes, says Purnomo, a vast improvement in operational efficiency and platform performance. Best of all, it

means the team of three engineers is now freed up almost entirely to focus on product innovation, such as real-time market analysis and product customization. This has translated into concrete results in terms of customer satisfaction and business growth.

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Thanks to a more resilient platform on Google Cloud that had zero outages since the migration, and a steady stream of new product features, the Bareksa app's Google Play Store rating has surged from 3.8 to 4.7. Platform improvements have enabled its AUM to increase fivefold.

The deployment of Cloud SQL as Bareksa's scalable relational database has further liberated the engineering team from mundane chores. Since moving to the Google Cloud managed solution, the team has achieved a 60% reduction in the time it spends configuring and managing databases for the Bareksa platform.

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Secure, reliable workflows to improve productivity with Google Workspace

The Bareksa team had been using a legacy productivity solution at the time of the Google Cloud migration, and one of the problems of this setup was regularly losing documents due to system crashes. The startup also lived in constant worry of security breaches hitting its on-premise computing networks.

For Purnomo, switching to a Google Cloud architecture provided the perfect opportunity to move to [Google Workspace](#)

(<https://workspace.google.com/>) as Bareksa's productivity solution. The natural compatibility of Workspace with Google Cloud, and its secure-by-design product range, gave Purnomo confidence that such a switch would unleash new creative synergies. His hunch was validated when the team immediately found an enhanced ability to optimize workflows and collaboration, while handling sensitive financial data in a safe and secure manner.

Not least, Bareksa was delighted to find that a persistent problem of losing at least two documents per month due to system failures disappeared completely, as Google Workspace automated the real-time saving of files to the cloud in Drive (https://workspace.google.com/intl/en_my/products/drive/) , Docs (<https://www.google.com/docs/about/>), and Sheets (<https://www.google.com/sheets/about/>).

"At first some employees were hesitant about leaving a productivity solution everyone was accustomed to," says Purnomo. "All of that dissipated overnight when the team discovered how easy and intuitive working on Google Workspace was, especially how important documents never went astray because everything is automatically consolidated in Drive."

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**Charting Indonesia's fintech future
with BigQuery data analytics**



Now that Bareksa has optimized operations with GKE and Google Workspace, the startup feels ready to turn to the next stage of its mission of Indonesian financial empowerment. This is to create a more curated financial product range for its customers through big data analytics.

To achieve the goal, Bareksa is adopting BigQuery (<https://cloud.google.com/bigquery>) to gain real-time machine learning insight into the AUM growth trajectories of its products. That will give it a granular readout of the services that are gaining the most traction with the public, and therefore enable it to tailor solutions to customer needs.

By deploying BigQuery's unlimited auto-scaling capabilities to crunch petabytes of data, Bareksa aims to obtain a finer understanding of customer patterns, and thus create more personalized services for each valued client.

"With BigQuery as our data analytics solution, we're confident we'll be able to invent the tools Indonesians need to make their own decisions about their future," says Purnomo. "That will enable us in

our broader mission to empower Indonesians in their journey to financial security."